

Problem

Design a series RLC resonant circuit with $\omega_0 = 40$ rad/s and $B = 10$ rad/s.

Step-by-step solution

Step 1 of 5

Consider the following data for the series RLC circuit:

Resonance frequency, $\omega_0 = 40$ rad/s.

Bandwidth of the circuit is $B = 10$ rad/s.

Step 2 of 5

Write the formula for the bandwidth.

$$B = \frac{R}{L} \dots\dots (1)$$

Here

L Inductor of the series RLC circuit

R resistance of the series RLC circuit

Rewrite the equation (1).

$$L = \frac{R}{B} \dots\dots (2)$$

Let us assume resistance, $R = 10 \Omega$.

Substitute 10 for B and 10 for R in equation (2).

$$L = \frac{10}{10} \text{ The inductor value for the desired circuit is } \boxed{1 \text{ H}}.$$

$$= 1 \text{ H}$$

Step 3 of 5

Write the quality factor expression in terms of bandwidth and resonant frequency.

$$Q = \frac{\omega_0}{B}$$

Substitute 40 for ω_0 and 10 for B in the above expression

$$Q = \frac{40}{10}$$

$$= 4$$

Therefore the quality factor of the desired circuit Q is $\boxed{4}$.

Step 4 of 5

Write the quality factor expression in terms of circuit components and resonant frequency.

$$Q = \frac{1}{\omega_0 CR}$$

Rewrite the expression.

$$C = \frac{1}{\omega_0 QR}$$

Substitute 40 for ω_0 , 10 for R and 4 for Q in the above expression.

$$\begin{aligned} C &= \frac{1}{(40)(4)(10)} \\ &= 625 \times 10^{-6} \text{ F} \\ &= 625 \mu\text{F} \end{aligned}$$

The capacitor value for the desired circuit is $625 \mu\text{F}$.

Step 5 of 5

Therefore the desired series RLC circuit is shown below.

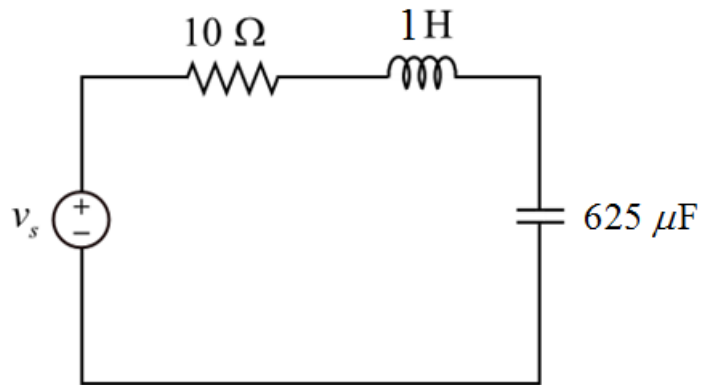


Figure 1